

School of Computer Science Engineering and Technology

Lab No. - 17

Total Marks—1

Course-B. Tech.	Type- Core
Course Code- CSET301	Course Name- Artificial Intelligence and Machine Learning
Year- 2025	Semester- Odd
Date- 23/09/2025	Batch- 2023-2027

CO-Mapping

	CO1	CO2	CO3	CO4	CO5	CO6
Q1			√			

Objective

The objective of this lab is to understand and implement the Naïve Bayes algorithm for text classification. Students will learn how to preprocess raw text into numerical features using CountVectorizer, train a Multinomial Naïve Bayes classifier, and evaluate its performance with accuracy and confusion matrix. This exercise demonstrates the practical application of Bayes' theorem in Natural Language Processing (NLP) tasks.

Dataset:

https://media.geeksforgeeks.org/wp-content/uploads/20250303132330481308/synthetic_text_data.csv

1. **Import Libraries** – Load the required Python libraries such as `pandas`, `numpy`, `scikit-learn` modules, and visualization tools.
2. **Load and Explore Dataset** – Read the text dataset, observe its structure, and separate the features (`text`) and target (`label`).
3. **Preprocess Text Data** – Convert raw text into numerical form using `CountVectorizer` to generate word count features.
4. **Train Naïve Bayes Classifier** – Apply the `MultinomialNB` model on the training data and build the classifier.
5. **Evaluate the Model** – Use accuracy score, confusion matrix, and visualization (heatmap) to assess model performance, and test predictions on unseen input.

Important links:

https://scikit-learn.org/stable/modules/naive_bayes.html

<https://www.datacamp.com/tutorial/naive-bayes-scikit-learn>

For further practice, explore how Naïve Bayes can be applied to spam detection using email/text data. Follow this tutorial

<https://medium.com/@afreenhossain/building-a-spam-detection-system-using-naive-bayes-and-countvectorizer-14702badf194>